International Institute of Information Technology, Hyderabad

(Deemed to be University)

Intro to VLSI and Embedded Systems

Quiz 1

Max. Time: 50 min **Max. Marks: 25**

Q1. Say we are planning to manufacture a new IC. We use a 200 mm diameter wafer, and our die size is 2 mm x 2 mm. We invested \$1 Million to make the design. The cost of the wafer is \$1000. The expected yield of the wafer is 99% and that of dies on the wafer is 95%. The packaging and test costs are \$0.10 per die. If we are planning the production of 10 Million chips, what is the cost per IC [8]? What should be our selling price for 50% profit [2]?

[8+2 = 10 marks]

Q2. We connect the inverting terminal of an open loop Opamp to an AC signal with peak voltage V_p. At the non-inverting terminal, we provide a variable DC voltage of V_r. This voltage can vary between $\pm V_p$. Provide an expression for the duty cycle of the output as a function of V_r [10].

Q3. We have a circuit running on $V_{dd} = 2$ V. The Ion for matched nMOS and pMOS is 1 mA. If the output capacitance is assumed to be 10 pF, what is the propagation delay for an inverter [3]? What is the frequency of oscillation for a 5-stage ring oscillator circuit using this technology [2]?

[3+2 = 5 marks]